

P-value and Confidence Interval

Researchers are responsible for reporting statistics truthfully and transparently, accurately representing evidence, acknowledging uncertainty, and supporting informed, meaningful decision-making (McShane *et al.*, 2024). However, as Greenland *et al.* (2016) note, these decisions are rarely simple or straightforward. As a result, pragmatic interpretations often emerge, relying on misleading binary "effect/no-effect" approaches that foster a false sense of certainty rather than reflecting genuine real-world importance.

Why is a p-value not always enough to report?

One of the key issues is the overreliance on the p-value, as highlighted by Greenland (2016) and Tijssen (2021). Both authors agree that while the p-value quantifies the compatibility of the observed data with the null hypothesis, it is insufficient on its own to draw meaningful conclusions. The p-value does not address other essential aspects, such as the magnitude, precision, or practical significance of an observed effect. It also overlooks uncertainty, bias, and the reliability of findings. The p-value is only one piece of the puzzle. Therefore, it is recommended to interpret the p-value alongside effect sizes, confidence intervals, and the broader research context, as emphasised by McShane *et al.* (2024), to fully address research questions and provide statistical interpretations that support effective decision-making.

Under what situations should confidence intervals not be reported?

Hazra (2017, p. 4126) emphasises that a confidence interval (CI) is "a range of values, calculated from sample observations, that is likely to contain the true population value with some degree of uncertainty." Greenland *et al.* (2016) and

Tijssen (2021) stress that this uncertainty, stemming from sampling variability, must be communicated honestly and transparently. Clearly quantifying uncertainty, estimating precision, and presenting the range of plausible effect sizes are crucial, regardless of whether the results are favourable. As both authors note, the CI provides complementary information about data quality, which may be limited by a small sample size, high variability, or low precision of estimates. Omitting the CI would conceal important evidence underpinning the analysis.

However, building on the guidance from the Office for National Statistics (n.d.), when the entire target population has been observed, and the objective is purely descriptive, confidence intervals are generally unnecessary because there is no sampling uncertainty to estimate. This raises the question of whether population inference is needed in such cases.

Overall, reflecting on current literature, I believe the focus should shift from when CIs may be omitted to consistently reporting CIs whenever inferential statistics are appropriate and population parameters are estimated. CIs provide transparent communication about an estimate's precision and uncertainty and should not be excluded due to wide intervals, non-significant results, or imperfect data. The limitation lies not with the CI itself. As Greenland et al. (2016) note, caution is warranted when interpreting CIs, as they do not reveal selection bias or model misspecification. Only carefully analysed statistical evidence—including p-values and CIs—combined with research context and practical relevance, can minimise misinterpretation and support effective clinical decision-making.

References

Greenland, S., *et al.* (2016) 'Statistical tests, P values, confidence intervals, and power: A guide to misinterpretations', *European Journal of Epidemiology*, 31(4), pp. 337–350. DOI: <https://doi.org/10.1007/s10654-016-0149-3>

Hazra, A. (2017) 'Using the confidence interval confidently', *Journal of Thoracic Disease*, 9(10), pp. 4125–4130. DOI: <https://doi.org/10.21037/jtd.2017.09.14>

McShane, B.B., *et al.* (2024) "'Statistical Significance" and Statistical Reporting: Moving Beyond Binary', *Journal of Marketing*, 88(3), pp. 1–19. DOI: <https://doi.org/10.1177/00222429231216910>

Office for National Statistics (n.d.) *Uncertainty and how we measure it for our surveys*. Available at: <https://www.ons.gov.uk/methodology/methodologytopicsandstatisticalconcepts/uncertaintyandhowwemeasureit#confidence-interval> (Accessed: 28 June 2026).

Schober, P.S., Bossers, S.M. and Schwarte, L.A. (2018) 'Statistical significance versus clinical importance of observed effect sizes: what do P values and confidence intervals really represent?', *Anesthesia & Analgesia*, 126(3), pp. 1068–1072. DOI: <https://doi.org/10.1213/ANE.0000000000002798>

Tijssen, J.G.P. (2021) 'More confidence intervals and fewer p values: A positive trend?', *Journal of the American College of Cardiology*, 77(12), pp. 1562–1563. DOI: <https://doi.org/10.1016/j.jacc.2021.02.004>